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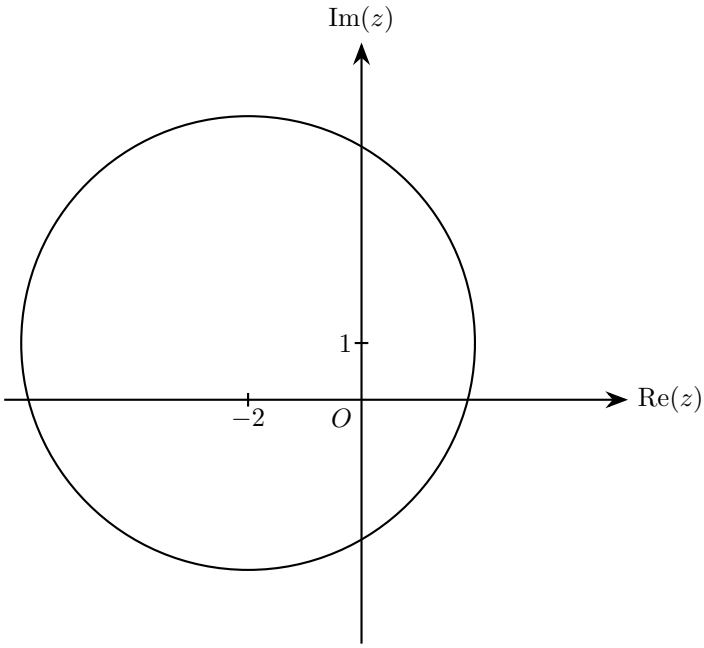


1. The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number $-2+\mathrm{i}$.

(a) Use set notation to define the locus of complex numbers, z, represented by points which lie on the circle. **[2]**

The locus L is defined by

$$L = \left\{ z \in \mathbb{C} \;\middle|\; |z-(-2+\mathrm{i})| = |z+1-3\mathrm{i}| \right\}$$

(b) On a copy of the Argand diagram, sketch and label the locus L. **[2]**

You are given that the locus

$$\left\{ z \in \mathbb{C} \;\middle|\; \arg(z+1-2\mathrm{i}) = \frac{2\pi}{3},\ \mathrm{Im}(z) = 5 \right\}$$

contains only one number.

(c) Find this number. **[2]**

Solution

(a) The circle has centre $-2+\mathrm{i}$ and radius 4.

So the locus is the set of complex numbers whose distance from $-2+\mathrm{i}$ is 4:

$$\{z \in \mathbb{C} \mid |z-(-2+\mathrm{i})| = 4\} = \{z \in \mathbb{C} \mid |z+2-\mathrm{i}| = 4\}$$

Hence the locus is $\{z \in \mathbb{C} \mid |z+2-\mathrm{i}| = 4\}$.

(b) Let $z = x+\mathrm{i}y$, where $x, y \in \mathbb{R}$.

Then

$$|z-(2+\mathrm{i})| = |z+1-3\mathrm{i}|$$

gives

$$|(x-2)+\mathrm{i}(y-1)| = |(x+1)+\mathrm{i}(y-3)|$$

Squaring both sides:

$$(x-2)^2 + (y-1)^2 = (x+1)^2 + (y-3)^2$$

Now expand:

$$\begin{aligned} x^2 - 4x + 4 + y^2 - 2y + 1 &= x^2 + 2x + 1 + y^2 - 6y + 9 \\ -4x - 2y + 5 &= 2x - 6y + 10 \\ -6x + 4y - 5 &= 0 \\ 6x - 4y + 5 &= 0 \end{aligned}$$

So

$$L: \ 6x - 4y + 5 = 0$$

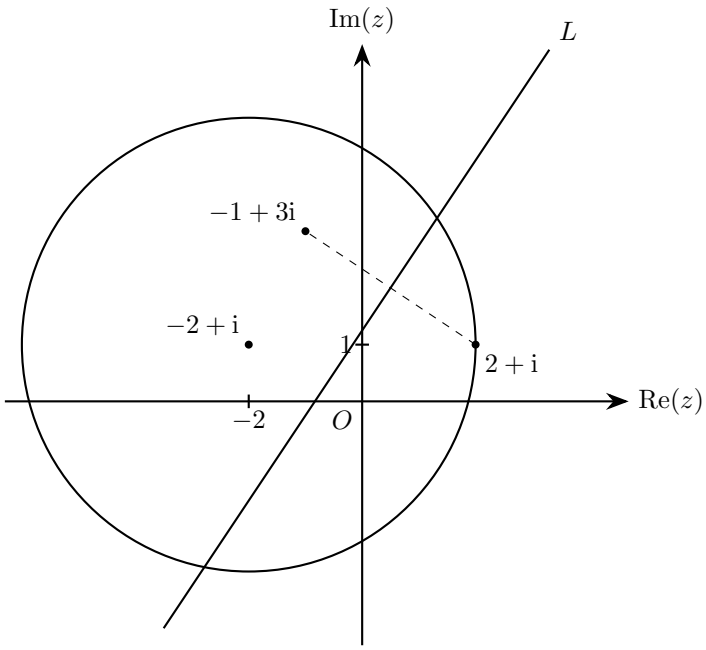
or equivalently

$$y = \frac{3}{2}x + \frac{5}{4}$$

This is a straight line. Since it is the set of points equidistant from $2 + \mathrm{i}$ and $-1 + 3\mathrm{i}$, it is the perpendicular bisector of the line segment joining those two points.

Its midpoint is

$$\left(\frac{2+(-1)}{2}, \frac{1+3}{2}\right) = \left(\frac{1}{2}, 2\right)$$



(c) Since $\mathrm{Im}(z) = 5$, write

$$z = x + 5\mathrm{i}$$

Then

$$z + 1 - 2\mathrm{i} = (x+1) + 3\mathrm{i}$$

We are given

$$\arg(z + 1 - 2\mathrm{i}) = \frac{2\pi}{3}$$

So the argument of $(x+1) + 3\mathrm{i}$ is $120°$, which is in the second quadrant. Therefore

$$\tan\frac{2\pi}{3} = \frac{3}{x+1}$$

Using $\tan \frac{2\pi}{3} = -\sqrt{3}$,

$$\frac{3}{x+1} = -\sqrt{3}$$

Hence

$$x + 1 = -\sqrt{3}$$

so

$$x = -1 - \sqrt{3}$$

Therefore

$$z = -1 - \sqrt{3} + 5\mathrm{i}$$

So the required number is $-1 - \sqrt{3} + 5\mathrm{i}$.

2. In an Argand diagram, the point P representing the complex number w lies on the locus defined by

$$\left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+\mathrm{i}) = \frac{\pi}{6}\right\}$$

You are given that $\mathrm{Im}(w) = 3$.

(a) Find w. **[2]**

The point P also lies on the locus defined by $\left\{z \in \mathbb{C} \;\middle|\; |z-2-11\mathrm{i}| = k\right\}$, where k is a constant.

(b) Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{z \in \mathbb{C} \;\middle|\; |z-2-11\mathrm{i}| = k\right\} \quad \text{and} \quad \left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+\mathrm{i}) = \frac{\pi}{6}\right\}$$

[7]

Solution

(a) Let

$$w = x + 3\mathrm{i}$$

since $\mathrm{Im}(w) = 3$.
Then

$$w - 2 + \mathrm{i} = (x-2) + 4\mathrm{i}$$

Given

$$\arg(w - 2 + \mathrm{i}) = \frac{\pi}{6}$$

so

$$\tan\frac{\pi}{6} = \frac{4}{x-2}$$

Hence

$$\frac{1}{\sqrt{3}} = \frac{4}{x-2}$$

so

$$x - 2 = 4\sqrt{3}$$

and therefore

$$x = 2 + 4\sqrt{3}$$

Thus

$$w = 2 + 4\sqrt{3} + 3\mathrm{i}$$

So the required complex number is $2 + 4\sqrt{3} + 3\mathrm{i}$.

(b) From part (a),

$$P \text{ represents } w = 2 + 4\sqrt{3} + 3\mathrm{i}$$

The second locus is

$$|z - 2 - 11\mathrm{i}| = k = |z - (2 + 11\mathrm{i})|$$

which is a circle with centre $2 + 11\mathrm{i}$.
Using the point P to find k,

$$\begin{aligned} k &= |w - (2 + 11\mathrm{i})| \\ &= \left|(2 + 4\sqrt{3} + 3\mathrm{i}) - (2 + 11\mathrm{i})\right| \\ &= \left|4\sqrt{3} - 8\mathrm{i}\right| \end{aligned}$$

So

$$\begin{aligned} k^2 &= (4\sqrt{3})^2 + (-8)^2 \\ &= 48 + 64 \\ &= 112 \end{aligned}$$

and hence

$$k = 4\sqrt{7}$$

Now let a general point of intersection be

$$z = x + y\mathrm{i}$$

Since

$$\arg(z - 2 + \mathrm{i}) = \frac{\pi}{6}$$

the point lies on the half-line starting at $(2, -1)$ with gradient

$$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

so its equation is

$$y + 1 = \frac{x-2}{\sqrt{3}}$$

that is,

$$y = \frac{x-2}{\sqrt{3}} - 1$$

The circle has equation

$$(x-2)^2 + (y-11)^2 = 112$$

Substitute $y = \dfrac{x-2}{\sqrt{3}} - 1$:

$$(x-2)^2 + \left(\frac{x-2}{\sqrt{3}} - 12\right)^2 = 112$$

Let

$$u = x - 2$$

Then

$$u^2 + \left(\frac{u}{\sqrt{3}} - 12\right)^2 = 112$$

Expanding,

$$\begin{aligned} u^2 + \left(\frac{u^2}{3} - 8\sqrt{3}\,u + 144\right) &= 112 \\ \frac{4u^2}{3} - 8\sqrt{3}\,u + 144 &= 112 \\ \frac{4u^2}{3} - 8\sqrt{3}\,u + 32 &= 0 \end{aligned}$$

Multiply by 3:

$$4u^2 - 24\sqrt{3}\,u + 96 = 0$$

Divide by 4:

$$u^2 - 6\sqrt{3}\,u + 24 = 0$$

Factorising,

$$(u - 4\sqrt{3})(u - 2\sqrt{3}) = 0$$

So

$$u = 4\sqrt{3} \quad \text{or} \quad u = 2\sqrt{3}$$

The value $u = 4\sqrt{3}$ gives the known point P from part (a), so the other intersection has

$$u = 2\sqrt{3}$$

Therefore

$$x = 2 + 2\sqrt{3}$$

and

$$y = \frac{2\sqrt{3}}{\sqrt{3}} - 1 = 2 - 1 = 1$$

Hence the other point represents

$$z = 2 + 2\sqrt{3} + \mathrm{i}$$

So the other complex number is $2 + 2\sqrt{3} + \mathrm{i}$.

3. On an Argand diagram, the locus of points satisfying $|z - 4 + 3\mathrm{i}| = 8$ is a circle.

(a) Write down

(i) the coordinates of the centre of this circle **[1]**

(ii) the radius of this circle **[1]**

(b) On an Argand diagram, shade the set of points defined by

$$|z - 4 + 3\mathrm{i}| \leq 8$$ **[1]**

(c) For the set of points defined in part (b), determine the maximum value of $|z|$. **[3]**

The set of points A is defined by

$$A = \left\{ z \in \mathbb{C} \;\middle|\; -\frac{\pi}{4} \leq \arg(z - 3 - 6\mathrm{i}) \leq \frac{3\pi}{4} \right\} \bigcap \left\{ z \in \mathbb{C} \;\middle|\; |z - 4 + 3\mathrm{i}| \leq 8 \right\}$$

(d) Determine the area of the region defined by A, giving your answer to 3 significant figures. **[4]**

Solution

(a) (i)

$$|z - 4 + 3\mathrm{i}| = |z - (4 - 3\mathrm{i})|$$

So the circle has centre $(4, -3)$.

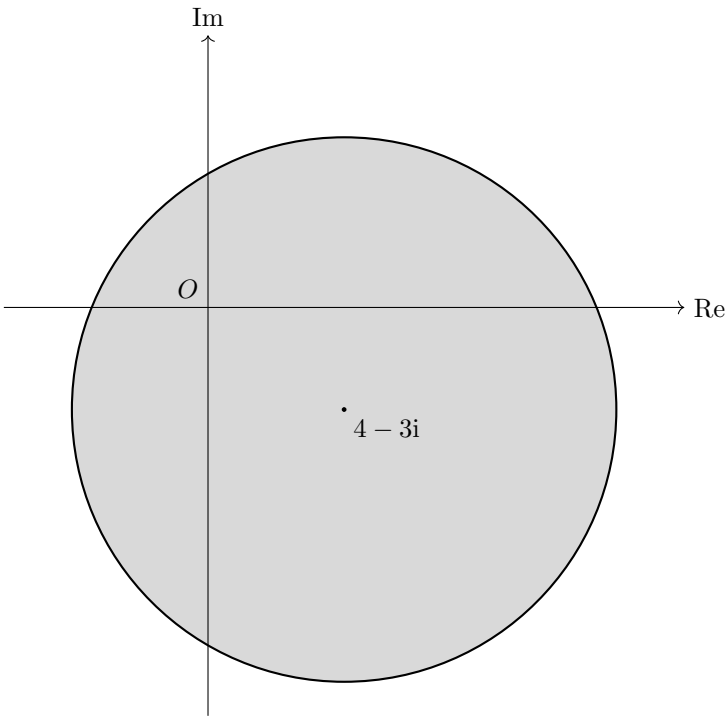
(ii) Comparing with the standard form $|z - a| = r$, the radius is 8.

(b) The inequality

$$|z - 4 + 3\mathrm{i}| \leq 8$$

means all points whose distance from $4 - 3\mathrm{i}$ is at most 8.

So we shade the interior and boundary of the circle with centre $(4, -3)$ and radius 8.



(c) The modulus $|z|$ is the distance from the origin.

For any point in the region,

$$\begin{aligned}|z| &= |(z-4+3\mathrm{i})+(4-3\mathrm{i})| \\ &\le |z-4+3\mathrm{i}|+|4-3\mathrm{i}| \\ &\le 8+\sqrt{4^2+(-3)^2} \\ &= 8+5 \\ &= 13\end{aligned}$$

Equality is possible at the point on the circle lying on the line from the origin through the centre.

Therefore the maximum value of $|z|$ is 13.

(d) Let $z = x + \mathrm{i}y$.

First consider the argument condition:

$$-\frac{\pi}{4} \le \arg(z-3-6\mathrm{i}) \le \frac{3\pi}{4}$$

Since

$$z-3-6\mathrm{i} = (x-3)+\mathrm{i}(y-6)$$

the boundary arguments $-\pi/4$ and $3\pi/4$ are opposite directions, so they form a straight line through $(3,6)$ with gradient -1.

Hence the boundary line is

$$y-6 = -(x-3)$$

so

$$x+y=9$$

To decide which side is needed, take the point $(3,7)$. Then

$$\arg\big((3+7\mathrm{i})-3-6\mathrm{i}\big) = \arg(\mathrm{i}) = \frac{\pi}{2}$$

which is in the given range, and $(3,7)$ satisfies $x+y=10$.

So the argument condition is the half-plane

$$x+y \ge 9$$

The circle has centre $C=(4,-3)$ and radius 8. Since

$$4+(-3)=1<9$$

the centre lies on the opposite side of the line, so the region A is the minor segment cut off by the line $x+y=9$.

The perpendicular distance from C to the line $x+y=9$ is

$$d = \frac{|4+(-3)-9|}{\sqrt{1^2+1^2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

Let P and Q be the points where the line meets the circle, and let M be the midpoint of chord PQ. Then $CM \perp PQ$, so in right-angled triangle CMP,

$$\cos\angle PCM = \frac{CM}{CP} = \frac{4\sqrt{2}}{8} = \frac{\sqrt{2}}{2}$$

Hence

$$\angle PCM = \frac{\pi}{4}$$

so the angle subtended by the chord is

$$\angle PCQ = 2\times\frac{\pi}{4} = \frac{\pi}{2}$$

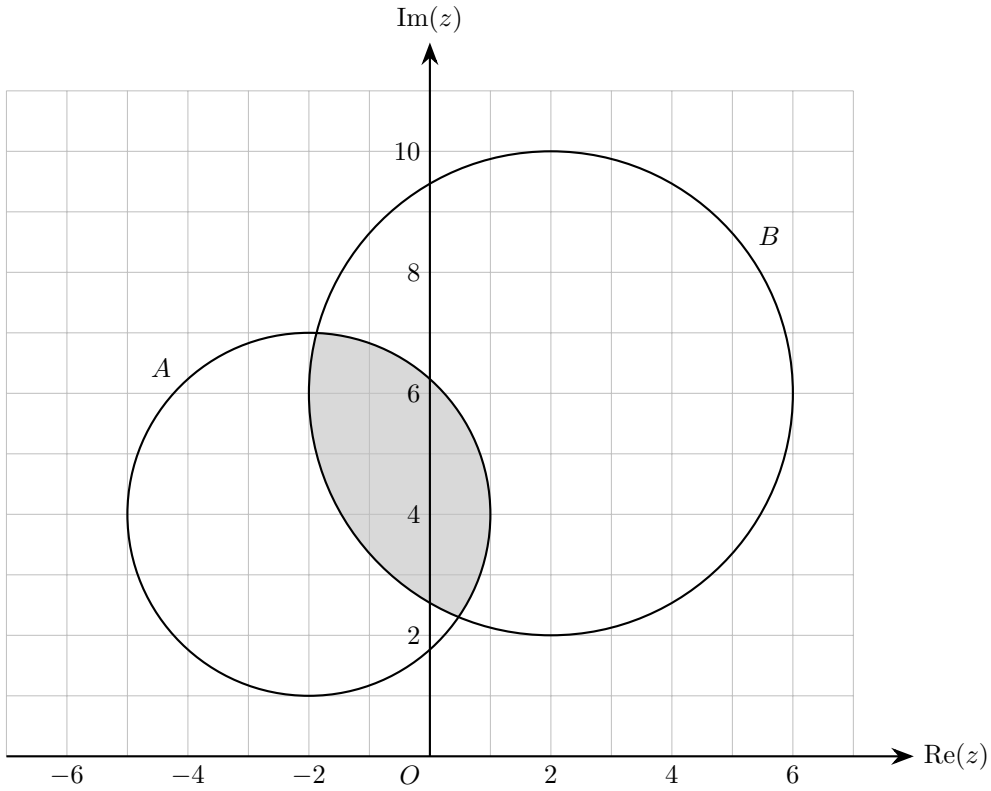
Therefore

$$\begin{aligned}\text{Area}(A) &= \text{area of sector } PCQ - \text{area of } \triangle PCQ \\ &= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta \\ &= \frac{1}{2}\cdot 8^2\cdot\frac{\pi}{2} - \frac{1}{2}\cdot 8^2\cdot\sin\frac{\pi}{2} \\ &= 16\pi - 32\end{aligned}$$

So the area is

$$16\pi - 32 \approx 18.3$$

to 3 significant figures.



4. **(a)** The diagram above shows an Argand diagram.

(i) Write down the equation of the locus represented by the circumference of circle A. **[3]**

(ii) Write down the two inequalities that define the shaded region, not including either circumference. **[3]**

(b) (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z + 2 - \mathrm{i}) < \frac{\pi}{4}$$ **[3]**

(ii) Determine whether the point $1 + 5\mathrm{i}$ lies within this region. **[3]**

Solution

(a) (i) For a circle with centre z_0 and radius r, the locus is

$$|z - z_0| = r$$

Circle A has centre $-2 + 4\mathrm{i}$ and radius 3, so

$$|z - (-2 + 4\mathrm{i})| = 3$$

hence

$$|z + 2 - 4\mathrm{i}| = 3$$

So the equation of the circumference of circle A is $|z + 2 - 4\mathrm{i}| = 3$.

(ii) The shaded region is the part inside both circles, and neither circumference is included.
Inside circle A:

$$|z + 2 - 4\mathrm{i}| < 3$$

Circle B has centre $2 + 6\mathrm{i}$ and radius 4, so inside circle B:

$$|z - (2 + 6\mathrm{i})| < 4$$

So the two inequalities are

$$|z + 2 - 4\mathrm{i}| < 3 \qquad \text{and} \qquad |z - (2 + 6\mathrm{i})| < 4$$

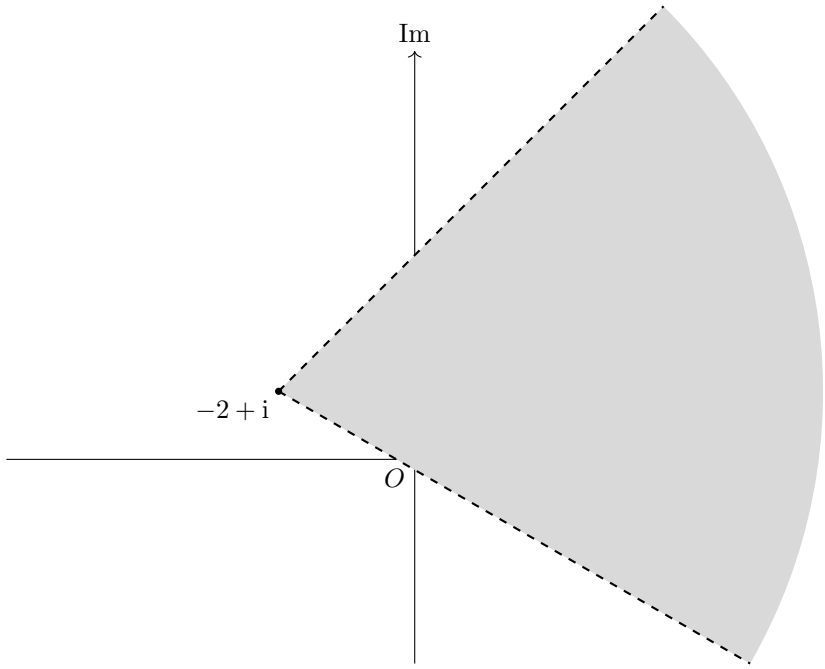
(b) (i) We have

$$z + 2 - \mathrm{i} = z - (-2 + \mathrm{i})$$

so the argument is measured from the point $-2 + \mathrm{i}$.
The region

$$-\frac{\pi}{6} < \arg(z + 2 - \mathrm{i}) < \frac{\pi}{4}$$

is therefore the open sector with vertex at $-2 + \mathrm{i}$ between the rays making angles $-\frac{\pi}{6}$ and $\frac{\pi}{4}$ with the positive real axis.
Since the inequalities are strict, both boundary rays are excluded.



(ii) For the point $z = 1 + 5\mathrm{i}$,

$$z + 2 - \mathrm{i} = (1 + 5\mathrm{i}) + 2 - \mathrm{i} = 3 + 4\mathrm{i}$$

So

$$\arg(z + 2 - \mathrm{i}) = \arg(3 + 4\mathrm{i}) = \tan^{-1}\left(\frac{4}{3}\right) \approx 0.927$$

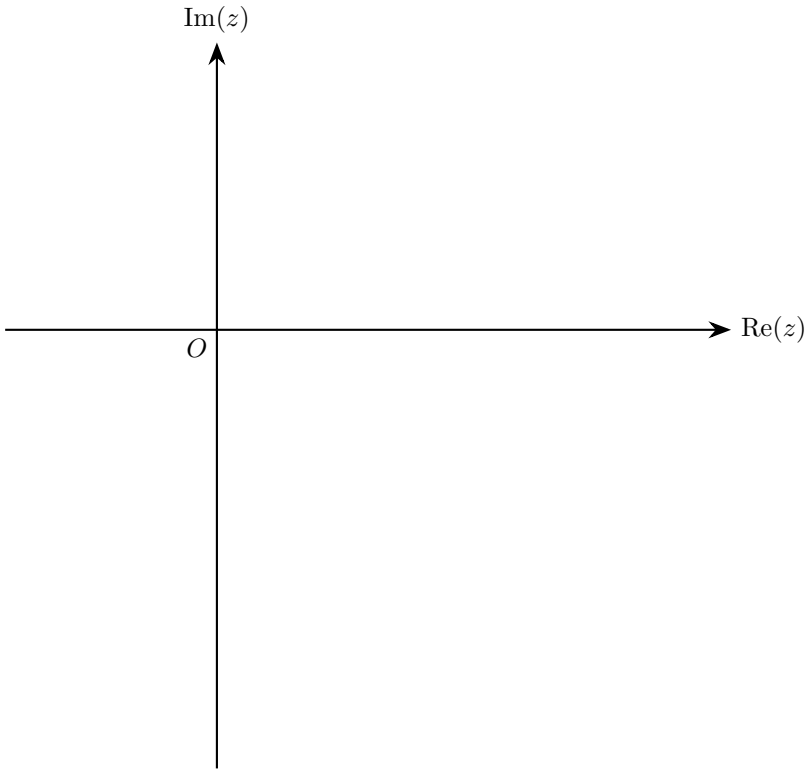
But

$$\frac{\pi}{4} \approx 0.785$$

Since

$$0.927 > \frac{\pi}{4}$$

the argument is too large, so $1 + 5\mathrm{i}$ does not lie in the region.
Therefore, the point $1 + 5\mathrm{i}$ is not in the region.



5. **(a)** Find the modulus of the complex number $2 + 2\sqrt{3}\mathrm{i}$, giving your answer as an integer. **[2]**

(b) The locus of points, L, satisfies the equation $|z - (2 - 2\sqrt{3}\mathrm{i})| = 2$.

(i) Sketch the locus L on the Argand diagram below. **[3]**

(ii) The complex number w lies on L so that $-\pi < \arg w \leq \pi$. Find the greatest possible value of $\arg w$, giving your answer in terms of π. **[2]**

(c) Solve the equation $z^4 = 2+2\sqrt{3}\mathrm{i}$, giving your answers in the form $re^{\mathrm{i}\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[5]**

Solution

(a) The modulus is

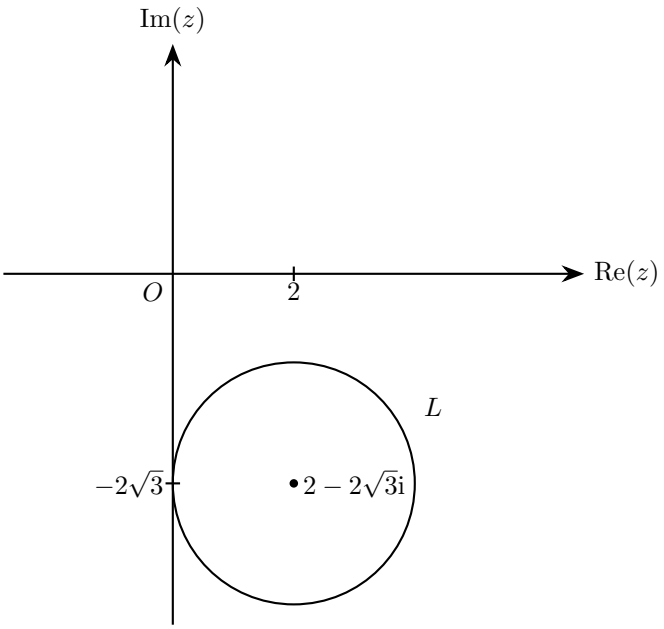
$$|2 + 2\sqrt{3}\mathrm{i}| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

So the modulus is 4.

(b) **(i)** The equation

$$|z - (2 - 2\sqrt{3}\mathrm{i})| = 2$$

represents all points a distance 2 from the point $2 - 2\sqrt{3}\mathrm{i}$.
So L is a circle with centre $(2, -2\sqrt{3})$ and radius 2.



(ii) Let C be the centre of the circle, so

$$C = 2 - 2\sqrt{3}\mathrm{i}$$

First find OC:

$$OC = |2 - 2\sqrt{3}\mathrm{i}| = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4+12} = 4$$

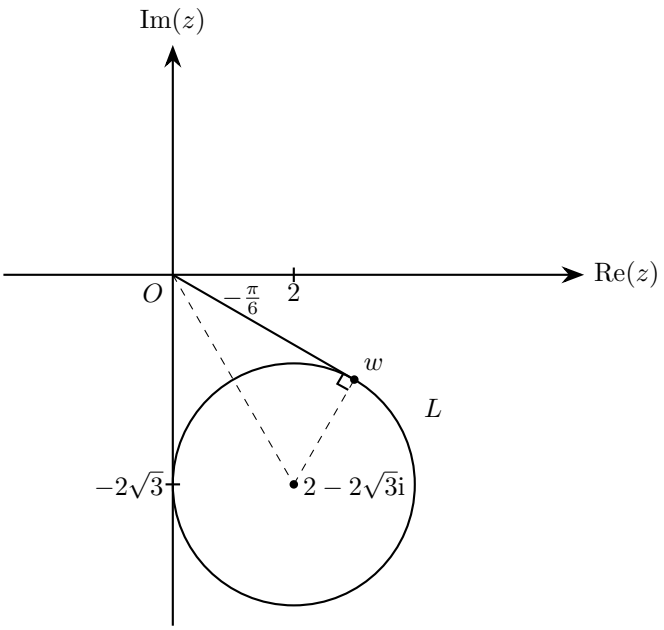
Also,

$$\arg C = -\frac{\pi}{3}$$

because C is in the fourth quadrant and

$$\tan|\arg C| = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

The greatest possible value of $\arg w$ occurs when the line from the origin to the circle is a tangent, touching the circle at w.



Then triangle OCW is right-angled at W, with

$$CW = 2, \qquad OC = 4$$

Let $\angle COW = \alpha$. Then

$$\sin\alpha = \frac{CW}{OC} = \frac{2}{4} = \frac{1}{2}$$

so

$$\alpha = \frac{\pi}{6}$$

Since this tangent is above the line OC,

$$\arg w = \arg C + \alpha = -\frac{\pi}{3} + \frac{\pi}{6} = -\frac{\pi}{6}$$

Therefore the greatest possible value of $\arg w$ is $-\frac{\pi}{6}$.

(c) From part (a),

$$|2 + 2\sqrt{3}\mathrm{i}| = 4$$

and since $2 + 2\sqrt{3}\mathrm{i}$ lies in the first quadrant,

$$\arg(2 + 2\sqrt{3}\mathrm{i}) = \frac{\pi}{3}$$

So

$$2 + 2\sqrt{3}\mathrm{i} = 4e^{\mathrm{i}\pi/3}$$

We solve

$$z^4 = 4e^{\mathrm{i}\pi/3}$$

Let $z = re^{\mathrm{i}\theta}$. Then

$$z^4 = r^4 e^{\mathrm{i}4\theta}$$

so

$$r^4 e^{\mathrm{i}4\theta} = 4e^{\mathrm{i}(\pi/3+2k\pi)} \qquad (k \in \mathbb{Z})$$

Equating moduli gives

$$r^4 = 4$$

so, with $r > 0$,

$$r = 4^{1/4} = \sqrt{2}$$

Equating arguments gives

$$4\theta = \frac{\pi}{3} + 2k\pi$$

hence

$$\theta = \frac{\pi}{12} + \frac{k\pi}{2}$$

The four distinct roots come from $k = 0, 1, 2, 3$:

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad \frac{13\pi}{12}, \quad \frac{19\pi}{12}$$

Writing these in the range $-\pi < \theta \leq \pi$,

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad -\frac{11\pi}{12}, \quad -\frac{5\pi}{12}$$

Therefore

$$z = \sqrt{2}e^{\mathrm{i}\pi/12}, \qquad \sqrt{2}e^{\mathrm{i}7\pi/12}, \qquad \sqrt{2}e^{-\mathrm{i}11\pi/12}, \qquad \sqrt{2}e^{-\mathrm{i}5\pi/12}$$

These are the four solutions.

6. The point P represents a complex number z on an Argand diagram such that

$$|z-(1+\mathrm{i})| = \frac{1}{2}|z-(4+\mathrm{i})|$$

(a) Show that, as z varies, the locus of P is a circle, and give the coordinates of the centre and the radius of the circle. **[5]**

The point Q represents a complex number z on an Argand diagram such that

$$|z-(2+3\mathrm{i})| = |z+2-\mathrm{i}|$$

(b) Sketch, on the same Argand diagram, the locus of P and the locus of Q as z varies. **[5]**

(c) On your diagram shade the region which satisfies

$$|z-(1+\mathrm{i})| \leq \frac{1}{2}|z-(4+\mathrm{i})| \quad \text{and} \quad |z-(2+3\mathrm{i})| \geq |z+2-\mathrm{i}|$$

[2]

Solution

(a) Let

$$z = x + \mathrm{i}y$$

Then

$$|z-(1+\mathrm{i})| = \sqrt{(x-1)^2+(y-1)^2}$$

and

$$|z-(4+\mathrm{i})| = \sqrt{(x-4)^2+(y-1)^2}$$

So the given condition becomes

$$\sqrt{(x-1)^2+(y-1)^2} = \frac{1}{2}\sqrt{(x-4)^2+(y-1)^2}$$

Squaring both sides,

$$4\big((x-1)^2+(y-1)^2\big) = (x-4)^2+(y-1)^2$$

Expand:

$$\begin{aligned} 4\big(x^2-2x+1+y^2-2y+1\big) &= x^2-8x+16+y^2-2y+1 \\ 4x^2-8x+4+4y^2-8y+4 &= x^2-8x+16+y^2-2y+1 \end{aligned}$$

Bring everything to one side:

$$\begin{aligned} 3x^2+3y^2-6y-9 &= 0 \\ x^2+y^2-2y-3 &= 0 \end{aligned}$$

Complete the square in y:

$$\begin{aligned} x^2+(y^2-2y+1)-4 &= 0 \\ x^2+(y-1)^2 &= 4 \end{aligned}$$

This is the equation of a circle.

Hence the locus is the circle with centre $(0,1)$ and radius 2.

(b) For the second locus, let again $z = x + \mathrm{i}y$.

Then

$$|z - (2 + 3\mathrm{i})| = \sqrt{(x-2)^2 + (y-3)^2}$$

and

$$|z + 2 - \mathrm{i}| = |z - (-2 + \mathrm{i})| = \sqrt{(x+2)^2 + (y-1)^2}$$

So

$$\sqrt{(x-2)^2 + (y-3)^2} = \sqrt{(x+2)^2 + (y-1)^2}$$

Squaring both sides,

$$(x-2)^2 + (y-3)^2 = (x+2)^2 + (y-1)^2$$

Expand and simplify:

$$\begin{aligned} x^2 - 4x + 4 + y^2 - 6y + 9 &= x^2 + 4x + 4 + y^2 - 2y + 1 \\ -4x - 6y + 13 &= 4x - 2y + 5 \\ -8x - 4y + 8 &= 0 \\ 2x + y - 2 &= 0 \end{aligned}$$

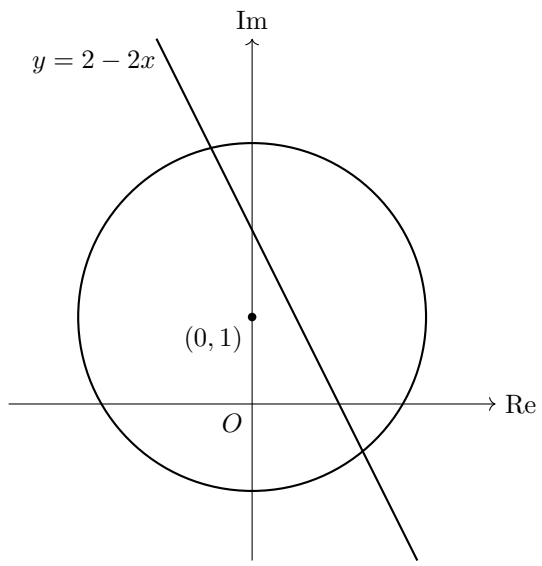
So the locus of Q is the straight line

$$y = 2 - 2x$$

This is the perpendicular bisector of the line segment joining $(2, 3)$ and $(-2, 1)$.

On the same Argand diagram, we sketch:

- the circle $x^2 + (y-1)^2 = 4$, centre $(0, 1)$, radius 2,
- the line $y = 2 - 2x$.



(c) From part (a),

$$|z - (1 + \mathrm{i})| \leq \frac{1}{2}|z - (4 + \mathrm{i})|$$

gives

$$x^2 + (y-1)^2 \leq 4$$

so the point lies inside or on the circle.

From part (b),

$$|z - (2 + 3\mathrm{i})| \geq |z + 2 - \mathrm{i}|$$

gives

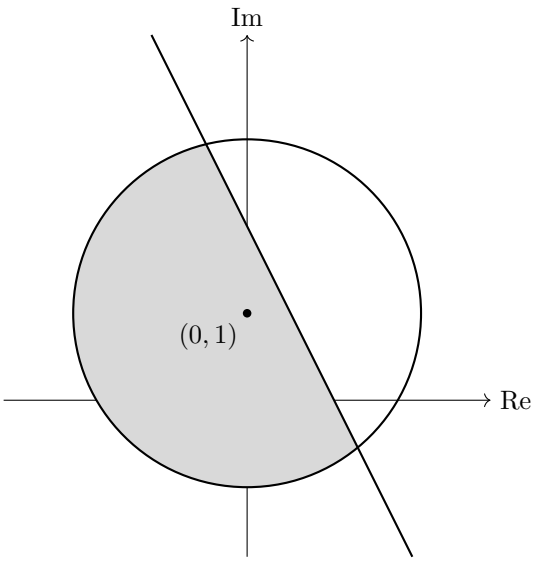
$$\begin{aligned}(x-2)^2+(y-3)^2 &\geq (x+2)^2+(y-1)^2\\ -8x-4y+8 &\geq 0\\ 2x+y-2 &\leq 0\\ y &\leq 2-2x\end{aligned}$$

So the point lies on or below the line $y = 2 - 2x$.

Therefore the required region is the intersection of:

- the inside of the circle $x^2 + (y-1)^2 = 4$,
- the half-plane $y \leq 2 - 2x$.

The shaded region is shown below.



So the region is inside the circle and on/below the line.

7. On an Argand diagram, the point P representing the complex number z lies on the locus

$$|z - (2 - \mathrm{i})| = 3$$

The point Q representing the complex number w lies on the locus

$$|w + 1 - 11\mathrm{i}| = 3|w + 1 - 3\mathrm{i}|$$

Find the exact coordinates of the points common to these two loci. **[7]**

Solution

Let a point common to both loci have affix

$$u = x + y\mathrm{i}$$

so its coordinates are (x, y).
For the locus of P,

$$|u - (2 - \mathrm{i})| = 3$$

Now

$$u - (2 - \mathrm{i}) = (x - 2) + (y + 1)\mathrm{i}$$

so

$$|(x - 2) + (y + 1)\mathrm{i}| = 3$$

Using $|a + b\mathrm{i}| = \sqrt{a^2 + b^2}$,

$$(x - 2)^2 + (y + 1)^2 = 9$$

For the locus of Q,

$$|u + 1 - 11\mathrm{i}| = 3|u + 1 - 3\mathrm{i}|$$

Now

$$u + 1 - 11\mathrm{i} = (x + 1) + (y - 11)\mathrm{i}, \qquad u + 1 - 3\mathrm{i} = (x + 1) + (y - 3)\mathrm{i}$$

So

$$|(x + 1) + (y - 11)\mathrm{i}| = 3|(x + 1) + (y - 3)\mathrm{i}|$$

Squaring both sides gives

$$(x + 1)^2 + (y - 11)^2 = 9\big((x + 1)^2 + (y - 3)^2\big)$$

Expand and simplify:

$$\begin{aligned} x^2 + 2x + 1 + y^2 - 22y + 121 &= 9x^2 + 18x + 9 + 9y^2 - 54y + 81 \\ 0 &= 8x^2 + 16x + 8 + 8y^2 - 32y - 32 \\ 0 &= x^2 + 2x + y^2 - 4y - 4 \\ 0 &= (x + 1)^2 + (y - 2)^2 - 9 \end{aligned}$$

Hence the second locus is

$$(x + 1)^2 + (y - 2)^2 = 9$$

So the common points satisfy the simultaneous equations

$$(x - 2)^2 + (y + 1)^2 = 9$$

and

$$(x + 1)^2 + (y - 2)^2 = 9$$

Subtracting one equation from the other:

$$\begin{aligned} (x - 2)^2 + (y + 1)^2 &= (x + 1)^2 + (y - 2)^2 \\ x^2 - 4x + 4 + y^2 + 2y + 1 &= x^2 + 2x + 1 + y^2 - 4y + 4 \\ -6x + 6y &= 0 \\ y &= x \end{aligned}$$

Substitute $y = x$ into $(x-2)^2 + (y+1)^2 = 9$:

$$\begin{aligned}(x-2)^2 + (x+1)^2 &= 9\\ x^2 - 4x + 4 + x^2 + 2x + 1 &= 9\\ 2x^2 - 2x - 4 &= 0\\ x^2 - x - 2 &= 0\\ (x-2)(x+1) &= 0\end{aligned}$$

So

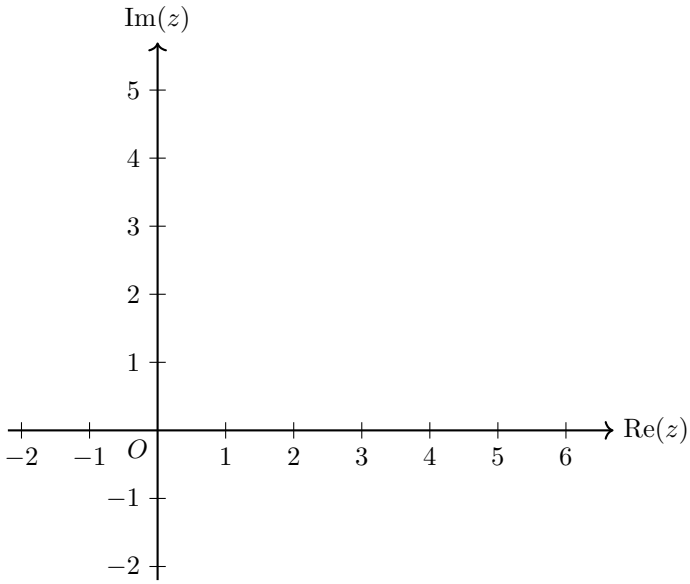
$$x = 2 \quad \text{or} \quad x = -1$$

and since $y = x$,

$$y = 2 \quad \text{or} \quad y = -1$$

Therefore the exact coordinates of the common points are

$$(2, 2) \quad \text{and} \quad (-1, -1)$$



8. **(a)** Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (3 + \mathrm{i})| = 1$$ **[1]**

(b) There is a unique complex number w that satisfies both

$$|w - (3 + \mathrm{i})| = 1 \quad \text{and} \quad \arg(w) = \alpha$$

where α is a constant such that $0 < \alpha < \pi$.

(i) Find the value of α. **[2]**

(ii) Find w and express it in the form $r(\cos\theta + \mathrm{i}\sin\theta)$. Give each of r and θ to two significant figures. **[4]**

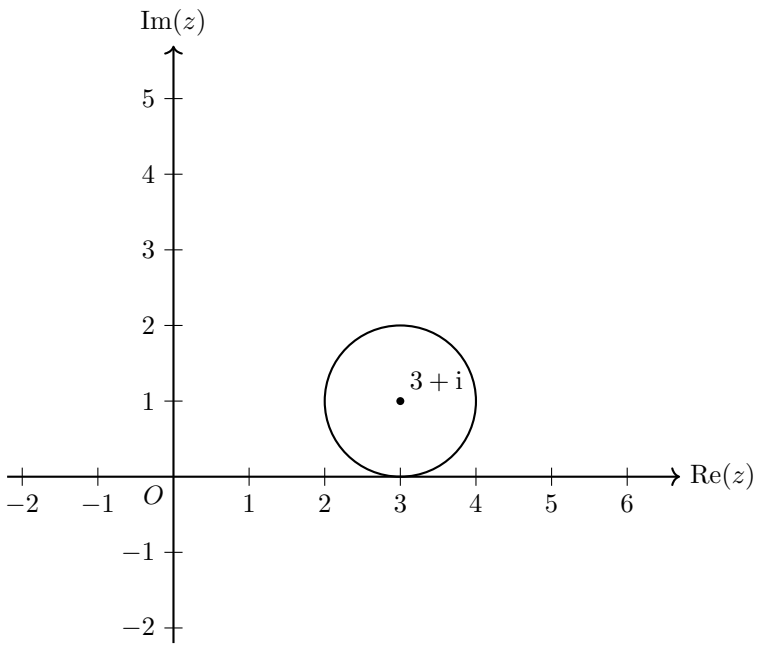
Solution

(a) The locus

$$|z - (3 + \mathrm{i})| = 1$$

is the set of points whose distance from $3 + \mathrm{i}$ is 1.

So it is a circle with centre $(3, 1)$ and radius 1.



(b) (i) If $\arg(w) = \alpha$, then w lies on a straight line through the origin, so its equation is

$$y = mx$$

where

$$m = \tan\alpha$$

For there to be a unique point w satisfying both conditions, this line must touch the circle at exactly one point, so it must be a tangent.
The circle has centre $(3, 1)$ and radius 1. The distance from $(3, 1)$ to the line $y = mx$, written as $mx - y = 0$, is

$$\frac{|3m - 1|}{\sqrt{m^2 + 1}}$$

Since this distance equals the radius,

$$\frac{|3m - 1|}{\sqrt{m^2 + 1}} = 1$$

Squaring both sides,

$$\begin{aligned}(3m - 1)^2 &= m^2 + 1\\ 9m^2 - 6m + 1 &= m^2 + 1\\ 8m^2 - 6m &= 0\\ 2m(4m - 3) &= 0\end{aligned}$$

So

$$m = 0 \quad \text{or} \quad m = \frac{3}{4}$$

If $m = 0$, then $\alpha = 0$, but the question states $0 < \alpha < \pi$, so this is not allowed.
Hence

$$\tan\alpha = \frac{3}{4}$$

so

$$\alpha = \arctan\left(\frac{3}{4}\right) \approx 0.64$$

Therefore, $\alpha = \arctan\left(\frac{3}{4}\right) \approx 0.64$ rad.

(ii) Using the value of m, the tangent line is

$$y = \frac{3}{4}x$$

The circle equation is

$$(x - 3)^2 + (y - 1)^2 = 1$$

Substituting $y = \frac{3}{4}x$,

$$\begin{aligned}(x - 3)^2 + \left(\frac{3}{4}x - 1\right)^2 &= 1\\ x^2 - 6x + 9 + \frac{9}{16}x^2 - \frac{3}{2}x + 1 &= 1\\ \frac{25}{16}x^2 - \frac{15}{2}x + 9 &= 0\end{aligned}$$

Multiplying by 16,

$$25x^2 - 120x + 144 = 0$$

Factorising,

$$(5x - 12)^2 = 0$$

So

$$x = \frac{12}{5}$$

Then

$$y = \frac{3}{4} \cdot \frac{12}{5} = \frac{9}{5}$$

Hence

$$w = \frac{12}{5} + \frac{9}{5}\mathrm{i}$$

Now find its modulus:

$$\begin{aligned} |w| &= \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2} \\ &= \sqrt{\frac{144}{25} + \frac{81}{25}} \\ &= \sqrt{\frac{225}{25}} \\ &= 3 \end{aligned}$$

Its argument is

$$\theta = \arg(w) = \alpha = \arctan\left(\frac{3}{4}\right) \approx 0.64$$

So in modulus-argument form,

$$w = 3(\cos 0.64 + \mathrm{i} \sin 0.64)$$

To two significant figures,

$$r = 3.0, \qquad \theta = 0.64$$

Therefore,

$$w = \frac{12}{5} + \frac{9}{5}\mathrm{i} = 3(\cos 0.64 + \mathrm{i} \sin 0.64)$$

9. The locus of points $z = x + \mathrm{i}y$ that satisfy

$$\arg\left(\frac{z-(2+\mathrm{i})}{z-(6+5\mathrm{i})}\right) = \frac{\pi}{4}$$

is an arc of a circle C.

(a) On an Argand diagram sketch the locus of z. **[2]**

(b) Explain why the centre of C lies on the line $y = 7 - x$. **[1]**

(c) Determine the radius of C. **[2]**

(d) Determine the coordinates of the centre of C. **[2]**

Solution

(a) Let

$$A = 2 + \mathrm{i} \quad \text{and} \quad B = 6 + 5\mathrm{i}$$

so $A = (2, 1)$ and $B = (6, 5)$.

The condition

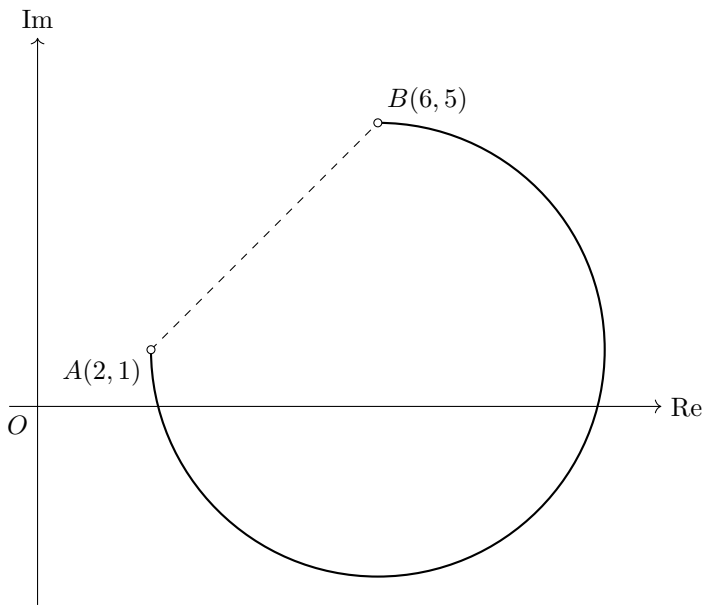
$$\arg\left(\frac{z-A}{z-B}\right) = \frac{\pi}{4}$$

means

$$\arg(z - A) - \arg(z - B) = \frac{\pi}{4}$$

so the angle between the lines ZA and ZB is $\frac{\pi}{4}$.

Hence the locus is an arc of a circle through A and B, and the required arc is the one on the side below/right of the line segment AB. Also, A and B are excluded because the expression is undefined there.



So the sketch is the arc through $(2, 1)$ and $(6, 5)$ on the side below/right of AB, with endpoints excluded.

(b) Since A and B lie on circle C, the centre of C is equidistant from A and B. Therefore it lies on the perpendicular bisector of AB.

The midpoint of AB is

$$\left(\frac{2+6}{2}, \frac{1+5}{2}\right) = (4, 3)$$

The slope of AB is

$$\frac{5-1}{6-2} = 1$$

so the perpendicular bisector has slope -1.

Its equation is

$$y - 3 = -1(x-4)$$
$$y = 7 - x$$

Therefore the centre lies on the line $y = 7 - x$.

(c) First find the length of chord AB:

$$|AB| = \sqrt{(6-2)^2 + (5-1)^2}$$

$$|AB| = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$$

The angle at the circumference is $\frac{\pi}{4}$, so the angle at the centre subtended by the same chord is

$$2 \times \frac{\pi}{4} = \frac{\pi}{2}$$

If the radius is r, then using the chord formula

$$|AB| = 2r \sin\left(\frac{1}{2} \times \frac{\pi}{2}\right)$$

$$4\sqrt{2} = 2r \sin\frac{\pi}{4} = 2r \cdot \frac{\sqrt{2}}{2} = r\sqrt{2}$$

So

$$r = 4$$

Therefore the radius of C is 4.

(d) Let M be the midpoint of AB, so $M = (4, 3)$.

From part (b), the centre lies on $y = 7 - x$, and from part (c) the radius is 4.

Also,

$$\frac{|AB|}{2} = 2\sqrt{2}$$

In the right-angled triangle formed by the centre, the midpoint of the chord, and one end of the chord,

$$OM = \sqrt{r^2 - \left(\frac{|AB|}{2}\right)^2}$$

$$OM = \sqrt{4^2 - (2\sqrt{2})^2} = \sqrt{16-8} = \sqrt{8} = 2\sqrt{2}$$

The line $y = 7 - x$ has direction vector

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

which has length $\sqrt{2}$. So moving a distance $2\sqrt{2}$ along this line means moving by

$$\pm 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Hence the two possible centres are

$$(4, 3) + (2, -2) = (6, 1)$$

and

$$(4, 3) + (-2, 2) = (2, 5)$$

To decide which is correct, test a point on the locus. Take $z = 10 + \mathrm{i}$:

$$\begin{aligned}\arg\left(\frac{(10+\mathrm{i})-(2+\mathrm{i})}{(10+\mathrm{i})-(6+5\mathrm{i})}\right) &= \arg\left(\frac{8}{4-4\mathrm{i}}\right)\\ &= \arg\left(\frac{8(4+4\mathrm{i})}{(4-4\mathrm{i})(4+4\mathrm{i})}\right)\\ &= \arg(1+\mathrm{i})\\ &= \frac{\pi}{4}\end{aligned}$$

So $10 + \mathrm{i}$ lies on the locus.

Now check which circle it lies on:

$$\text{distance from } (10,1) \text{ to } (6,1) = 4$$

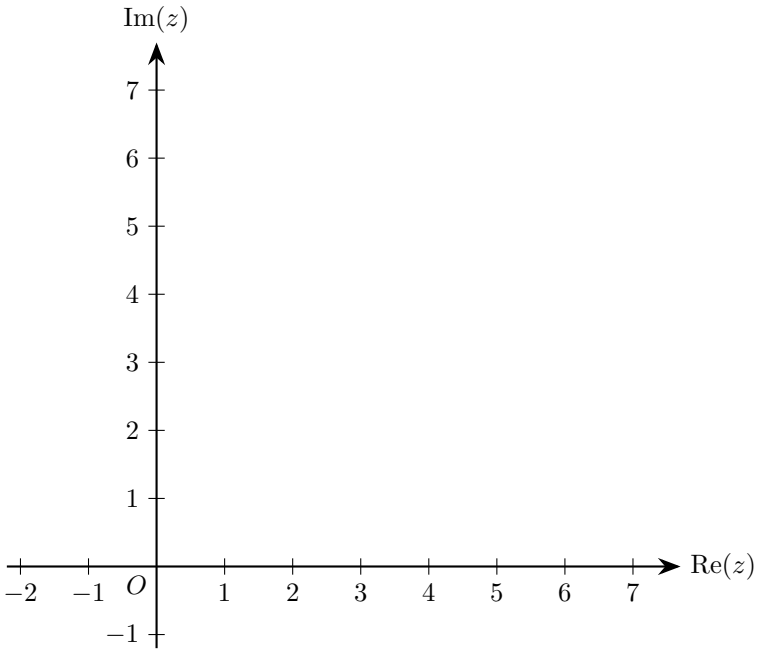
so it lies on the circle centred at $(6, 1)$.

But

$$\text{distance from } (10,1) \text{ to } (2,5) = \sqrt{(8)^2 + (-4)^2} = \sqrt{80} \neq 4$$

so it does not lie on the circle centred at $(2, 5)$.

Therefore the centre of C is $(6, 1)$.



10. **(a)** Sketch, on the Argand diagram below, the locus of points satisfying the equation

$$|z - (2 + 4\mathrm{i})| = 3$$ **[2]**

(b) Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4}$$ **[1]**

(c) For the complex number w, find the maximum value of $|w|$ such that

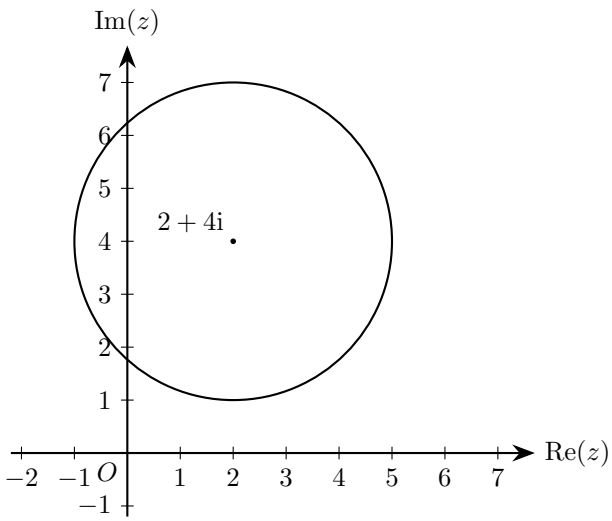
$$|w - (2 + 4\mathrm{i})| \leq 3 \qquad \text{and} \qquad 0 \leq \arg(w - 1) \leq \frac{\pi}{4}$$ **[3]**

Solution

(a) The locus

$$|z - (2 + 4\mathrm{i})| = 3$$

is a circle with centre at the point representing $2 + 4\mathrm{i}$ and radius 3



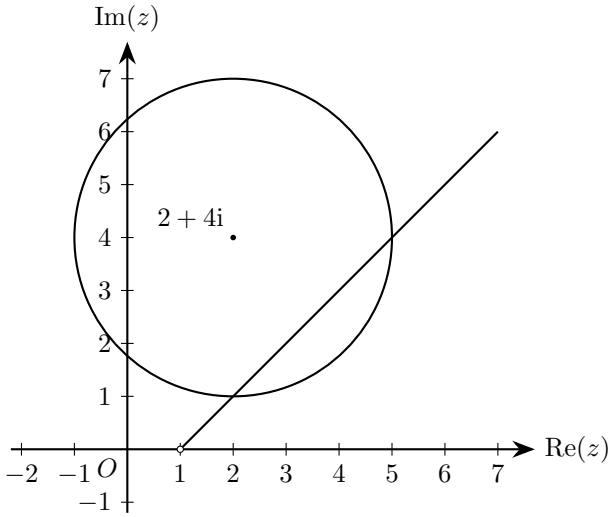
(b) The locus

$$\arg(z-1) = \frac{\pi}{4}$$

is the half-line starting at the point representing 1 on the real axis, making an angle of $\frac{\pi}{4}$ with the positive real axis

In Cartesian form this is

$$y = x - 1, \qquad x > 1$$



(c) Let

$$w = x + \mathrm{i}y$$

The condition

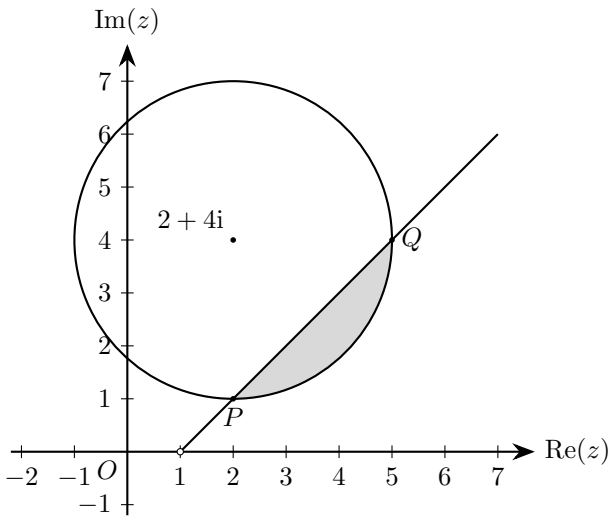
$$|w - (2 + 4\mathrm{i})| \leq 3$$

means that w lies inside or on the circle from part (a)

Also,

$$0 \leq \arg(w-1) \leq \frac{\pi}{4}$$

means that w lies in the region from the real axis up to the line $y = x - 1$



Since the circle has lowest point $(2, 1)$, every point in the circle has $y \geq 1$, so the condition $\arg(w-1) \geq 0$ is automatically satisfied

Therefore the required region is the part of the disc lying on or below the line

$$y = x - 1$$

First find where this line meets the circle:

$$\begin{aligned}(x-2)^2+(y-4)^2&=9\\(x-2)^2+(x-1-4)^2&=9\\(x-2)^2+(x-5)^2&=9\\x^2-4x+4+x^2-10x+25&=9\\2x^2-14x+20&=0\\x^2-7x+10&=0\\(x-2)(x-5)&=0\end{aligned}$$

so the intersection points are

$$(2,1) \quad \text{and} \quad (5,4)$$

Now use the circle inequality:

$$\begin{aligned}(x-2)^2+(y-4)^2&\leq 9\\x^2-4x+4+y^2-8y+16&\leq 9\\x^2+y^2&\leq 4x+8y-11\end{aligned}$$

But in the required region, $y \leq x-1$, so

$$\begin{aligned}|w|^2=x^2+y^2&\leq 4x+8y-11\\&\leq 4x+8(x-1)-11\\&=12x-19\end{aligned}$$

Also, inside the circle the greatest possible value of x is 5, so

$$|w|^2 \leq 12(5)-19=41$$

Equality is possible when $x=5$ and $y=x-1=4$, giving

$$w=5+4\mathrm{i}$$

Hence the maximum value of $|w|$ is

$$|w|_{\max}=\sqrt{41}$$

So the maximum value is $\sqrt{41}$, attained when $w=5+4\mathrm{i}$